SURENDAR L

Embedded Systems & IoT | AI-Integrated Electronics and Web | Web Development
Chennai, India | surendar7645321@gmail.com | linkedin.com/in/surendar-l-046bab343 | 9894520904 |
github.com/surendar050 | Surendar L. — ECE Engineer | Embedded Systems · IoT · AI

PROFESSIONAL SUMMARY

SEEKING FOR AN INTERNSHIP OPPORTUNITY.

B.Tech Electronics and Communication Engineering student at Pondicherry University with hands-on experience in embedded systems, IoT sensor integration, computer vision, and AI-integrated hardware projects. Proficient in Arduino microcontroller programming (Embedded C), YOLO-based real-time object detection, MQTT/BLE/REST communication protocols, and full-stack IoT mobile application development using React Native. Demonstrated ability to bridge hardware and software — from sensor data acquisition and signal conditioning to cloudconnected dashboards and AI-driven control systems. Seeking to contribute to ECE-focused roles involving embedded design, IoT product development, or AI/CV-integrated electronics systems.

TECHNICAL SKILLS

Embedded Systems	RapberryPi, Arduino ,ESP 32, Embedded C, Sensor Integration
IoT & Communication	MQTT, WebSockets, REST APIs, GSM/WiFi Device Provisioning, AWS IoT
Computer Vision & AI	YOLO (YOLOv5/v8) Object Detection, OpenCV, DeepFace (face recognition), Image Processing, Predictive Analytics, NLP .
Programming Languages	Python, Embedded C, JavaScript , React js , MERN stack ,
Design	PCB design (KiCAD) , Blender(3D Design)
Backend & Cloud	Node.js, Flask, Firebase, MongoDB, My SQL, Python scripting
Dev Tools & Platforms	Protius , Ansys EM Student , Blender, KiCAD ,WSL, Debian Linux ,Keil microvision , VS Code, Android Studio ,Git, GitHub , Arduino IDE, ESP32 IDE, Postman , Mysql Workbench , Mongo Compase , Wireshark

PROJECTS

1. AirSense – IoT Air Quality Monitoring Dashboard | *React Native · TypeScript · IoT · NLP Chatbot*

A complete IoT environmental sensing dashboard with real-time sensor data visualization and an embedded AI chatbot for natural language pollutant queries — built fully in React Native .

- Developed an embedded system using **ESP32** and **BME280 sensor** to measure real-time environmental parameters (temperature, humidity, and pressure)
- Implemented **I2C communication protocol** for efficient data exchange between microcontroller and sensor
- Engineered end-to-end IoT data pipeline: CSV sensor data ingestion via react-native-fs and papaparse for raw pollutant stream processing
- Integrated rule-based NLP chatbot enabling natural language queries on sensor data (highest/lowest/average pollutant levels)

GitHub: github.com/surendar050/AIRSENSE-app

2. Density-Based Adaptive Traffic Light System | *Python · YOLO · OpenCV · Arduino · Embedded C*

An intelligent traffic management system combining YOLO-based computer vision with Arduino microcontroller hardware, replacing static timer-based signals with dynamic, density-aware control.

- Implemented real-time vehicle detection using YOLOv5 on live camera feeds to count vehicle density per lane
- Programmed Arduino microcontroller in Embedded C to dynamically adjust traffic light durations based on detected vehicle counts
- Developed Python serial communication pipeline between YOLO inference engine and Arduino hardware over UART

GitHub: github.com/surendar050/Density-based-trafic-light-

3. Face Recognition System | *Python · DeepFace · OpenCV · REST API*

An automated AI-powered attendance system using deep learning facial recognition with OpenCV for real-time video-based identity verification.

- Built real-time face detection and multi-identity recognition pipeline using DeepFace deep learning library with OpenCV live video capture
- Implemented confidence-scoring system for reliable identity matching; REST API-ready backend architecture for IoT/mobile integration
- Demonstrates AI model API integration directly applicable to embedded AI in security, access control, and smart electronics systems

GitHub: github.com/surendar050/Face-Recognition

4. FastRider – Rapido like Mobile Application | *React Native · IoT · REST API · Firebase*

- Cross-platform React Native app with real-time data streaming , and Firebase cloud integration
- Implemented dynamic UI with real-time state management for high-frequency data updates — aligned with production architecture

GitHub: github.com/surendar050/fsatrider

EDUCATION

- **10th – IJMHSS –** State Board of Tami lNadu ***–83%*** **2021**
- **12th – IJMHSS –** State Board of Tamil Nadu ***–70%*** **2023**
- **B.Tech – Electronics and Communication Engineering – 7.31 CGPA** **2028 – Present**
  Pondicherry University, Puducherry

INTERNSHIP EXPERIENCE

Government Internship – Pondicherry Cyber Crime Unit 5/2025

- Assisted cybersecurity operations; gained exposure to digital forensics, network security monitoring, and threat analysis
- Applied understanding of secure communication protocols relevant to IoT security and cloudconnected embedded device safety

CERTIFICATIONS & TRAINING

- Front-End Development Certificate – Mindluster
- Cybersecurity Awareness – Government Cyber Crime Unit, Pondicherry
- Linux Administration and Shell Scripting